

Lecture 2

Ananya Roy

Lecturer

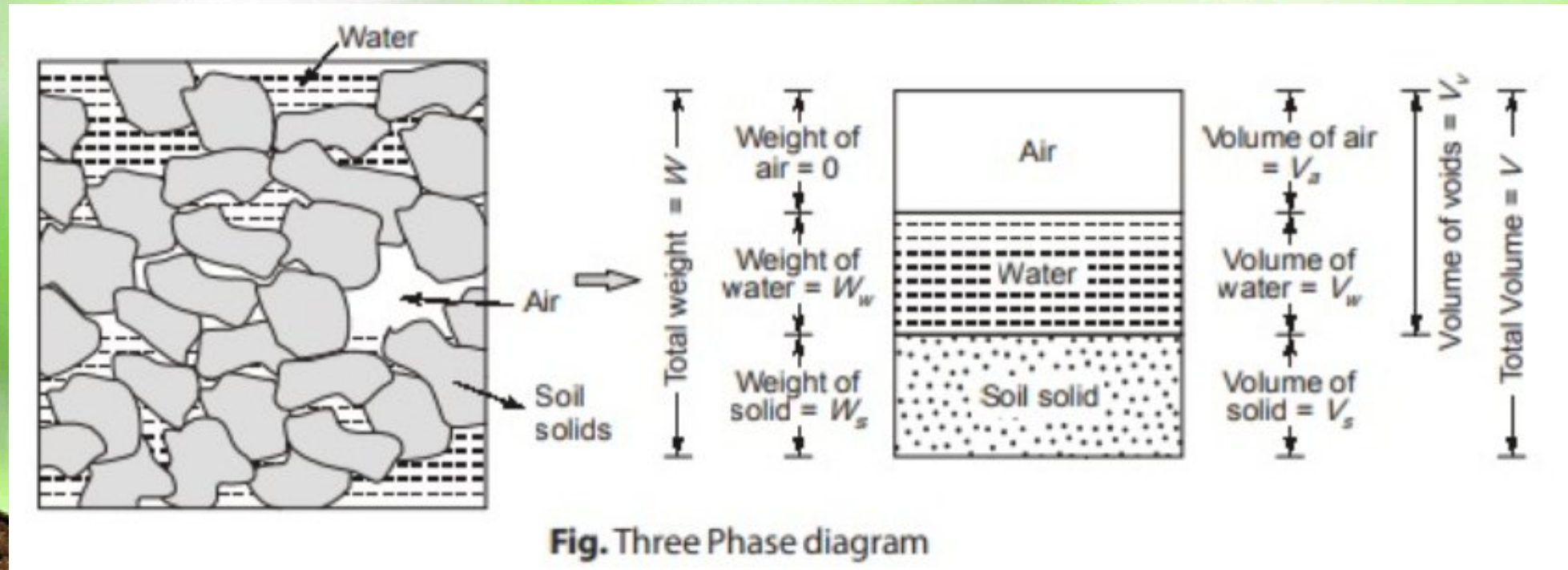
Environmental Science and Engineering

Jatiya Kabi Kazi Nazrul Islam University

Soil as Three phase system

- A soil mass consist of the solid particles and the voids in between them. These voids are filled with air or/and water.
- Soil mass, in general is a three phase system composed of solid, liquid and gaseous
- So there is a three phase system, but when the voids are only filled with air, or only filled with water then soil becomes a two phase system. When the voids are only filled with water, it is said to be **saturated**.
- Three phase system can be represented with a diagram as shown below.

Three phase diagram



In phase diagram, the weight of soil air is negligible compared to soil solid and water, thus it is assumed as zero. The diagrammatic representation of different soil phases is called phase diagram or block diagram.

Two phase diagram

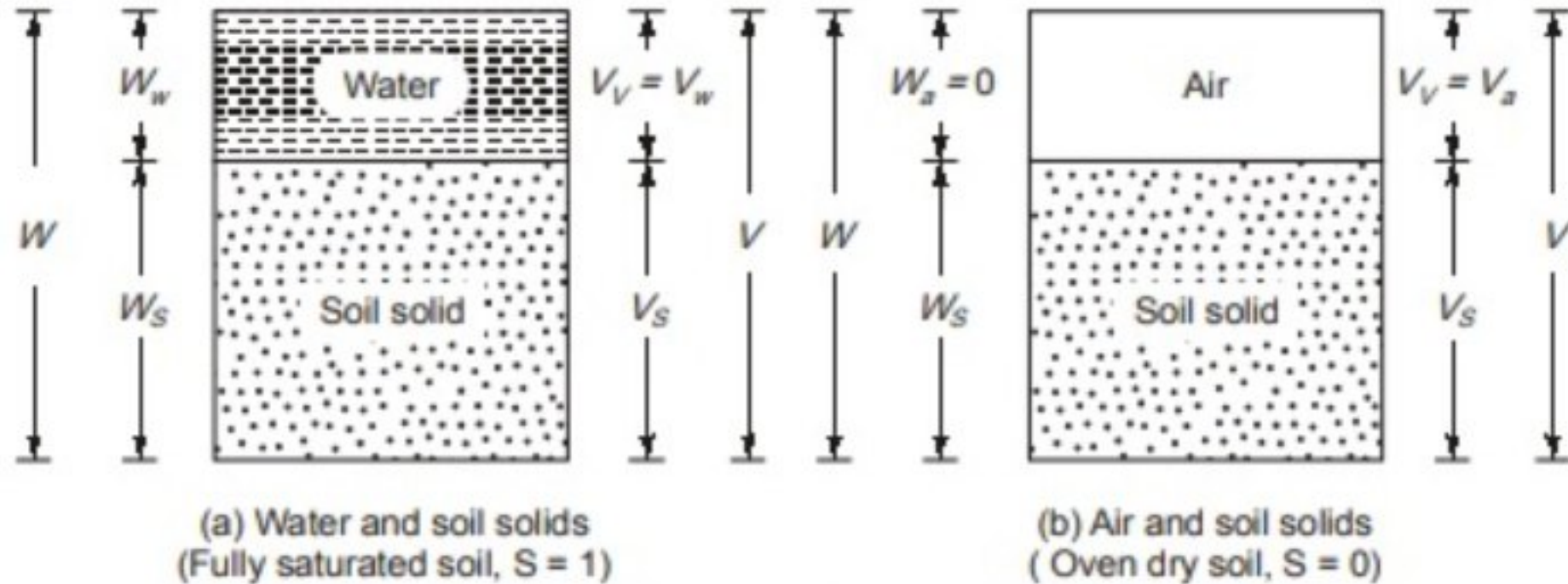


Fig. Two Phase diagram

Soil Phase relationship

Mass-Volume Relationship

In SI units, the mass M , is normally expressed in kg and the density ρ in kg/m³. Sometimes, the mass and densities are also expressed in g and g/cm³ or Mg and Mg/m³ respectively. The density of water ρ_w at 4 °C is exactly 1.00 g/cm³ (= 1000 kg/m³ = 1 Mg/m³). Since the variation in density is relatively small over the range of temperatures encountered in ordinary engineering practice, the density of water ρ_w at other temperatures may be taken the same as that at 4 °C. The volume is expressed either in cm³ or m³.

Weight-Volume Relationship

Unit weight or weight per unit volume is still the common measurement in geotechnical engineering practice. The density ρ , may be converted to unit weight, γ by using the relationship

$$\gamma = \rho g \quad \left[\text{Bulk (total) unit weight } \gamma = \frac{W}{V} = \frac{M \times g}{V} \right] \quad (3.1a)$$

The 'standard' value of g is 9.807 m/s² (= 9.81 m/s² for all practical purposes).

Conversion of Density of Water ρ_w to Unit Weight γ_w

From Eq. (3.1a)

$$\gamma_w = \rho_w g \quad (3.1b)$$

Substituting $\rho_w = 1000 \text{ kg/m}^3$ and $g = 9.81 \text{ m/s}^2$, we have

$$\gamma_w = 1000 \frac{\text{kg}}{\text{m}^3} \left(9.81 \frac{\text{m}}{\text{s}^2} \right) = 9810 \frac{\text{kg} \cdot \text{m}}{\text{m}^3 \text{s}^2} \quad (3.1c)$$

Since $1 \text{ N (newton)} = \frac{1 \text{ kg} \cdot \text{m}}{\text{s}^2}$, we have,

$$\gamma_w = 9810 \frac{\text{N}}{\text{m}^3} = 9.81 \frac{\text{kN}}{\text{m}^3} \quad (3.1d)$$

$$\text{or } \gamma_w = \left(1 \times \frac{\text{g}}{\text{cm}^3} \right) \times 9.81 = 9.81 \text{ kN/m}^3 \quad (3.1e)$$

In general, the unit weight of a soil mass may be obtained from the equation

$$\gamma = 9.81 \rho \text{ kN/m}^3 \quad (3.1f)$$

where in Eq. (3.1f), ρ is in g/cm^3 . For example, if a soil mass has a dry density, $\rho_d = 1.7 \text{ g/cm}^3$, the dry unit weight of the soil is

$$\gamma_d = 9.81 \times 1.7 = 16.68 \text{ kN/m}^3 \quad (3.1g)$$

Mass-Volume Ratio

Volumetric Ratios

There are three volumetric ratios that are very useful in geotechnical engineering and these can be determined directly from the phase diagram, Fig. 3.1.

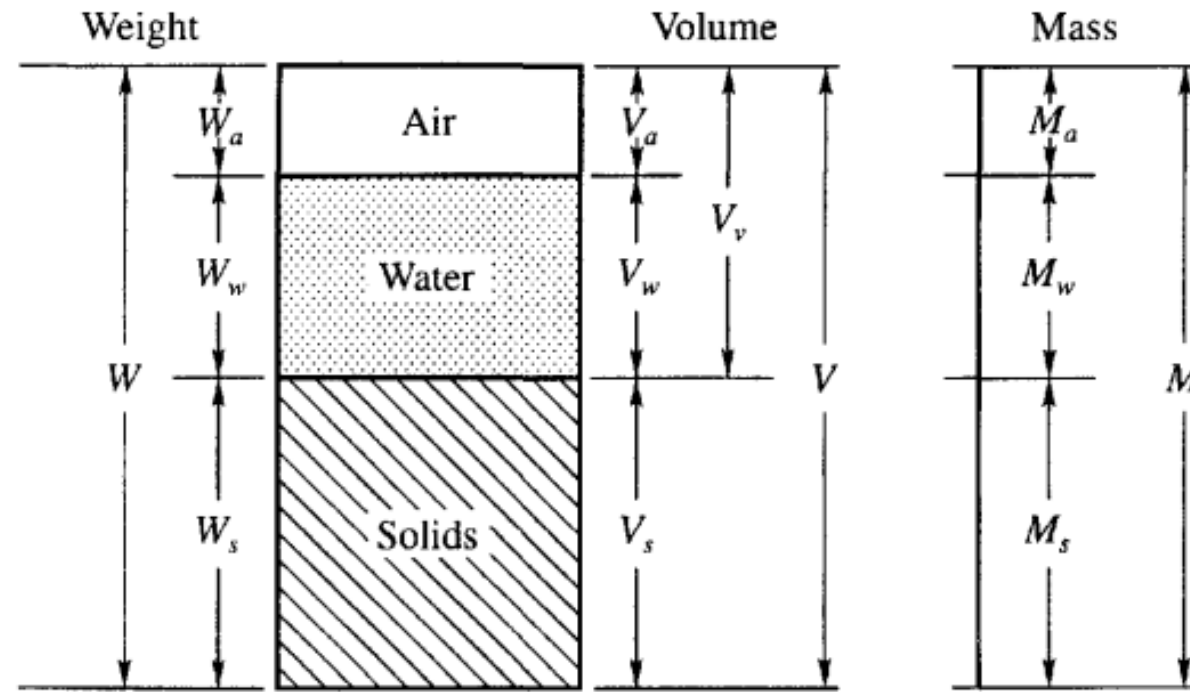


Figure 3.1 Block diagram—three phases of a soil element

1. The *void ratio*, e , is defined as

$$e = \frac{V_v}{V_s} \quad (3.2)$$

where, V_v = volume of voids, and V_s = volume of the solids.

The void ratio e is always expressed as a *decimal*.

2. The *porosity* n is defined as

$$n = \frac{V_v}{V} \times 100\% \quad (3.3)$$

where, V = total volume of the soil sample.

The *porosity* n is always expressed as a *percentage*.

3. The *degree of saturation* S is defined as

$$S = \frac{V_w}{V_v} \times 100\% \quad (3.4)$$

where, V_w = volume of water

It is always expressed as a *percentage*. When $S = 0\%$, the soil is completely dry, and when $S = 100\%$, the soil is fully saturated.

Water Content, w

The water content, w , of a soil mass is defined as the ratio of the mass of water, M_w , in the voids to the mass of solids, M_s , as

$$w = \frac{M_w}{M_s} \times 100 \quad (3.5)$$

The water content, which is usually expressed as a percentage, can range from zero (dry soil) to several hundred percent. The natural water content for most soils is well under 100%, but for the soils of volcanic origin (for example bentonite) it can range up to 500% or more.

Density

Another very useful concept in geotechnical engineering is *density* (or, unit weight) which is expressed as mass per unit volume. There are several commonly used densities. These may be defined as the total (or bulk), or moist density, ρ_t ; the dry density, ρ_d ; the saturated density, ρ_{sat} ; the density of the particles, solid density, ρ_s ; and density of water ρ_w . Each of these densities is defined as follows with respect to Fig. 3.1.

$$\text{Total density,} \quad \rho_t = \frac{M}{V} \quad (3.6)$$

Density Equations

Dry density, $\rho_d = \frac{M_s}{V}$

Saturated density, $\rho_{\text{sat}} = \frac{M}{V}$

for $S = 100\%$

Density of solids, $\rho_s = \frac{M_s}{V_s}$

Density of water, $\rho_w = \frac{M_w}{V_w}$

Specific Gravity

The specific gravity of a substance is defined as the ratio of its mass in air to the mass of an equal volume of water at reference temperature, 4 °C. The specific gravity of a mass of soil (including air, water and solids) is termed as bulk specific gravity G_m . It is expressed as

$$G_m = \frac{\rho_t}{\rho_w} = \frac{M}{V\rho_w} \quad (3.11)$$

The specific gravity of solids, G_s , (excluding air and water) is expressed by

$$G_s = \frac{\rho_s}{\rho_w} = \frac{M_s}{V_s\rho_w} \quad (3.12)$$

Interrelationships of Different Parameters

We can establish relationships between the different parameters defined by equations from (3.2) through (3.12). In order to develop the relationships, the block diagram Fig. 3.2 is made use of. Since the sectional area perpendicular to the plane of the paper is assumed as unity, the heights of the blocks will represent the volumes. The volume of solids may be represented as $V_s = 1$. When the soil is fully saturated, the voids are completely filled with water.

Relationship Between e and n (Fig. 3.2)

$$e = \frac{V_v}{V_s} = \frac{V_v}{1} = V_v, \quad n = \frac{V_v}{V} = \frac{e}{1+e}$$
$$\text{or } e = \frac{n}{1-n} \quad (3.13)$$

Relationship Between e , G_s and S

Case 1: When partially saturated ($S < 100\%$)

$$S = \frac{V_w}{V_v} = \frac{V_w}{e}; \quad \text{but } V_w = \frac{M_w}{\rho_w} = \frac{wM_s}{\rho_w} = \frac{wG_s V_s \rho_w}{\rho_w} = wG_s$$

$$\text{Therefore, } S = \frac{wG_s}{e} \quad \text{or } e = \frac{wG_s}{S}$$

Case 2: When saturated ($S = 100\%$)

From Eq. (3.14a), we have (for $S = 1$)

$$e = wG_s$$

Relationships Between Density ρ and Other Parameters

The density of soil can be expressed in terms of other parameters for cases of soil (1) partially saturated ($S < 100\%$); (2) fully saturated ($S = 100\%$); (3) Fully dry ($S = 0$); and (4) submerged.

Case 1: For $S < 100\%$

$$\rho_t = \frac{M}{V} = \frac{M_s(1+w)}{1+e} = \frac{G_s \rho_w (1+w)}{1+e} \quad (3.15)$$

From Eq. (3.14a) $w = eS/G_s$; substituting for w in Eq. (3.15), we have

$$\rho_t = \frac{\rho_w (G_s + eS)}{1+e} \quad (3.16)$$

Case 2: For $S = 100\%$

From Eq. (3.16)

$$\rho_t = \rho_{\text{sat}} = \frac{\rho_w (G_s + e)}{1+e} \quad (3.17)$$

Case 3: For $S = 0\%$

From Eq. (3.16)

$$\rho_t = \rho_d = \frac{\rho_w G_s}{1+e} \quad (3.18)$$

Example 3.1

A sample of wet silty clay soil has a mass of 126 kg. The following data were obtained from laboratory tests on the sample: Wet density, $\rho_t = 2.1 \text{ g/cm}^3$, $G = 2.7$, water content, $w = 15\%$.

Determine (i) dry density, ρ_d , (ii) porosity, (iii) void ratio, and (iv) degree of saturation.

Solution

Mass of sample $M = 126 \text{ kg}$.

$$\text{Volume } V = \frac{126}{2.1 \times 10^3} = 0.06 \text{ m}^3$$

$$\text{Now, } M_s + M_w = M, \text{ or } M_s + wM_s = M_s(1 + w) = M$$

$$\text{Therefore, } M_s = \frac{M}{1 + w} = \frac{126}{1.15} = 109.57 \text{ kg}; \quad M_w = M - M_s = 16.43 \text{ kg}$$

Now, $V_w = \frac{M_w}{\rho_w} = \frac{16.43}{1000} = 0.01643 \text{ m}^3;$

$$V_s = \frac{M_s}{G_s \rho_w} = \frac{109.57}{2.7 \times 1000} = 0.04058 \text{ m}^3;$$

$$V_v = V - V_s = 0.06000 - 0.04058 = 0.01942 \text{ m}^3.$$

(i) Dry density, $\rho_d = \frac{M_s}{V} = \frac{109.57}{0.06} = 1826.2 \text{ kg/m}^3$

(ii) Porosity, $n = \frac{V_v}{V} \times 100 = \frac{0.01942 \times 100}{0.06} = 32.37\%$

(iii) Void ratio, $e = \frac{V_v}{V_s} = \frac{0.01942}{0.04058} = 0.4786$

(iv) Degree of saturation, $S = \frac{V_w}{V_v} \times 100 = \frac{0.01643}{0.01942} \times 100 = 84.6\%$

Example 1: A soil has void ratio = 0.72, moisture content = 12% and $G_s = 2.72$. Determine its

- (a) Dry unit weight
- (b) Moist unit weight, and the
- (c) Amount of water to be added per m^3 to make it saturated.

Use $\gamma_w = 9.81 \text{ kN/m}^3$

Solution:

$$(a) \quad \gamma_d = \frac{G_s \cdot \gamma_w}{1+e} = \frac{2.72 \times 9.81}{1+0.72} = 15.51 \text{ kN/m}^3$$

$$(b) \quad \gamma = \gamma_d(1+w) \\ = \frac{1+0.12}{1+0.72} \times 2.72 \times 9.81 = 17.38 \text{ kN/m}^3$$

$$(c) \quad \gamma_{sat} = \frac{G_s + e}{1+e} \cdot \gamma_w \\ = \frac{2.72+0.72}{1+0.72} \times 9.81 = 19.62 \text{ kN/m}^3$$

Water to be added per m^3 to make the soil saturated

$$= \gamma_{sat} - \gamma = 19.62 - 17.38 = 2.24 \text{ kN}$$

Example 2: The dry density of a sand with porosity of 0.387 is 1600 kg/m^3 . Find the void ratio of the soil and the specific gravity of the soil solids. [Take $\gamma_w = 1000 \text{ kg/m}^3$]

$$n = 0.387$$

$$\gamma_d = 1600 \text{ kg/m}^3$$

Solution:

$$\text{(a) } e = \frac{n}{1-n} = \frac{0.387}{1-0.387} = 0.631$$

$$\text{(b) } \gamma_d = \frac{G_s \cdot \gamma_w}{1+e}$$

$$\therefore G_s = \frac{(1+e)}{\gamma_w} \cdot \gamma_d = \frac{1+0.631}{1000} \times 1600 = 2.61$$